EXPERIMENT 1

Analysis, Development and Verification of 8-bit and 16-bit Arithmetic Operations Using Assembly Language Program

a) Develop and Verification of (i) 8-Bit Addition and (ii) Subtraction Using 8-Bit Microprocessor Assembly Language Programming

b) Verification of (iii)16-Bit Mul and (iv) Division Using 8-Bit Microprocessor Assembly Language Programming

Objective:

To develop and verify **8-bit addition and subtraction** and **16-bit multiplication and division** operations using **8085 8-bit microprocessor assembly language programming**, and to analyze the obtained results.

Algorithm:

- Initialize the required memory locations and registers.
- Load the input data into the appropriate registers.
- Perform **8-bit addition** using the ADD instruction and store the result.
- Perform **8-bit subtraction** using the SUB instruction and store the result.
- Load the 16-bit operands into the required register pairs.
- Perform **16-bit multiplication** using repeated addition.
- Perform **division** using repeated subtraction.
- Store the quotient and remainder in the specified memory locations.
- Verify the obtained results with the expected arithmetic values.

a) (i) 8-bit Addition:

Code:

```
LDA 4500h
MOV B A
LDA 4501h
SUB B
STA 4502h
MVI A 00h
ACI 00h
STA 4503h
HLT
```

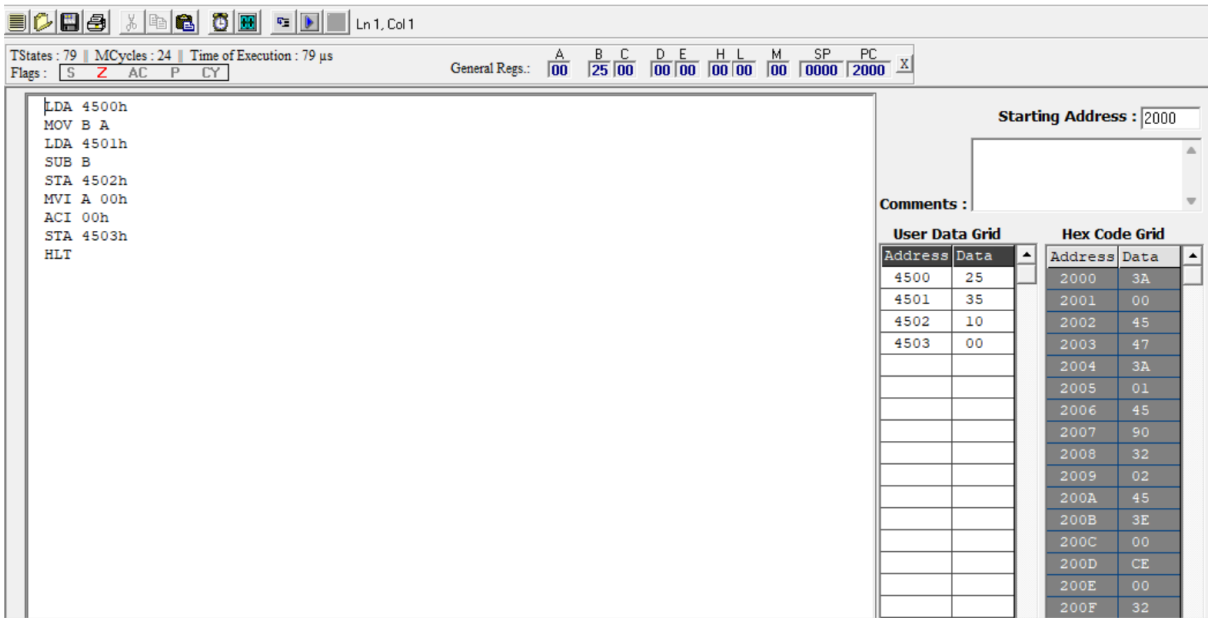
Input:

Input Address	Data
4500h	25h
4501h	35h

Output:

Output Address	Data
4502h	10h
4503h	00h

Output Screenshort:



(ii) 8-bit Subtraction:

Code:

```
LDA 4500h
MOV B A
LDA 4501h
SUB B
STA 4502h
MVI A 00h
ACI 00h
STA 4503h
HLT
```

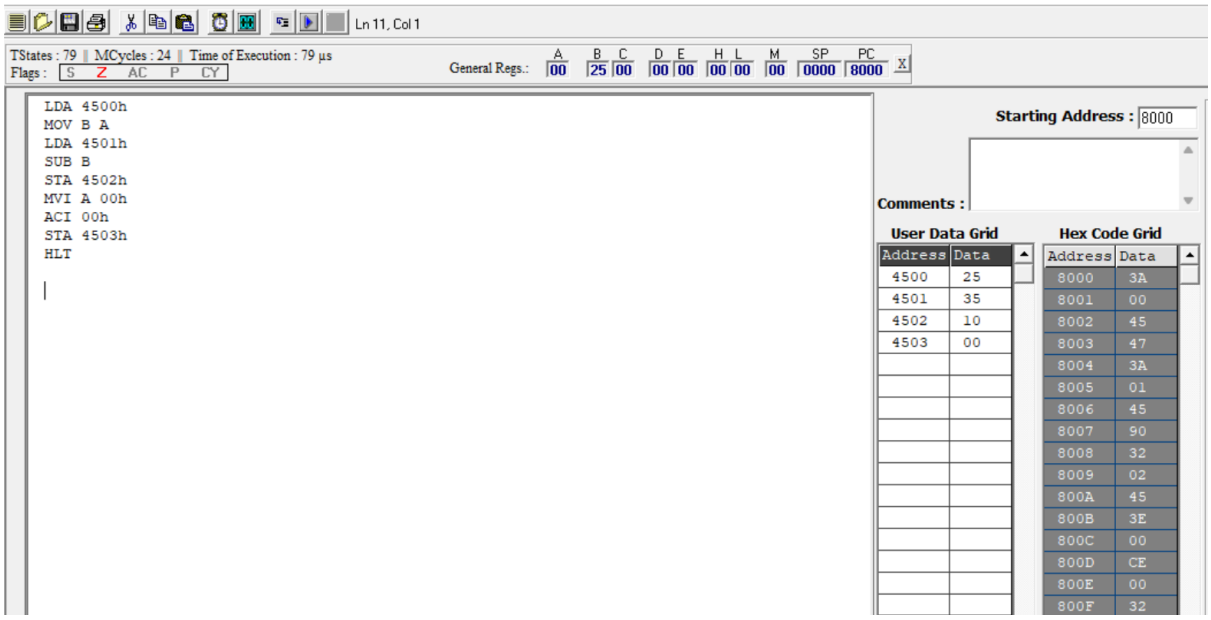
Input:

Input Address	Data
4500h	25h
4501h	35h

Output:

Output Address	Data
4502h	10h
4503h	00h

Output Screenshort:



b) (iii) 16-bit Multiplication Using 8-bit Processor

Code:

```
LXI H 0012h
MOV D H
MOV E L
LXI H 0003h
MOV B H
MOV C L
LXI H 0000h
LOOP: DAD D
DCR C
JNZ LOOP
SHLD 4502h
HLT
```

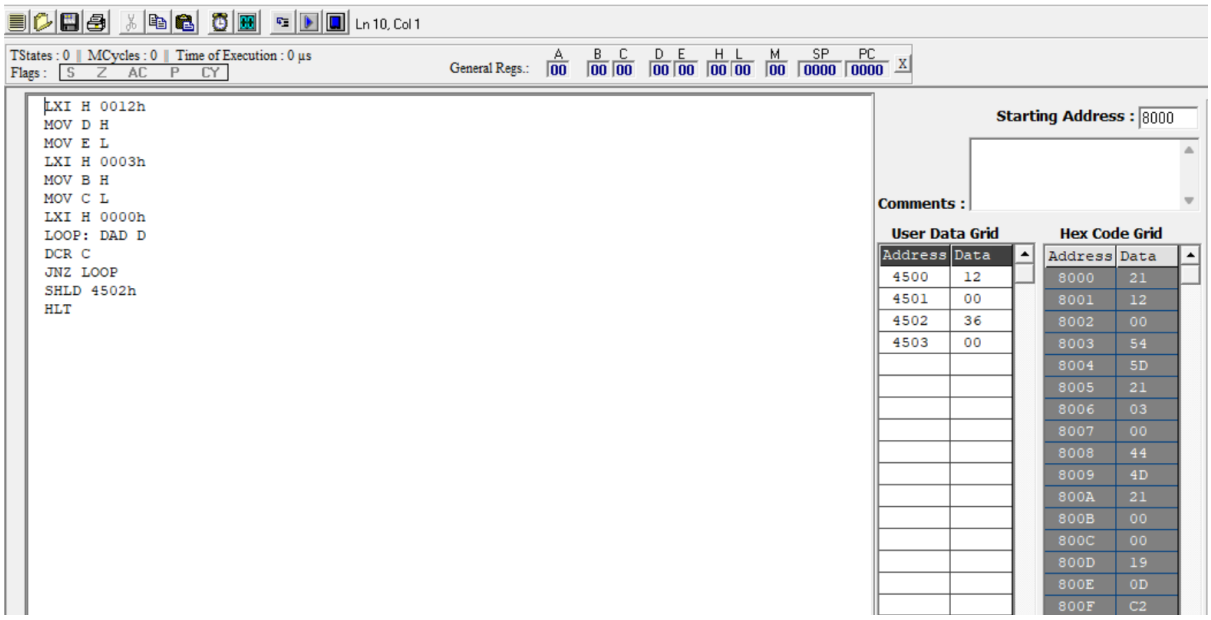
Input:

Input Address	Data
4500h	12h
4501h	00h

Output:

Output Address	Data
4502h	36h
4503h	00h

Output Screenshort:



(iv) 16-bit Division Using 8-bit Processor

Code:

```
LXI H 0012h
MOV D H
MOV E L
LXI H 0003h
MOV B H
MOV C L
LXI H 0000h
LOOP: MOV A E
SUB C
MOV E A
MOV A D
SBB B
MOV D A
JC DONE
INX H
JMP LOOP
DONE: SHLD 4502h
HLT
```

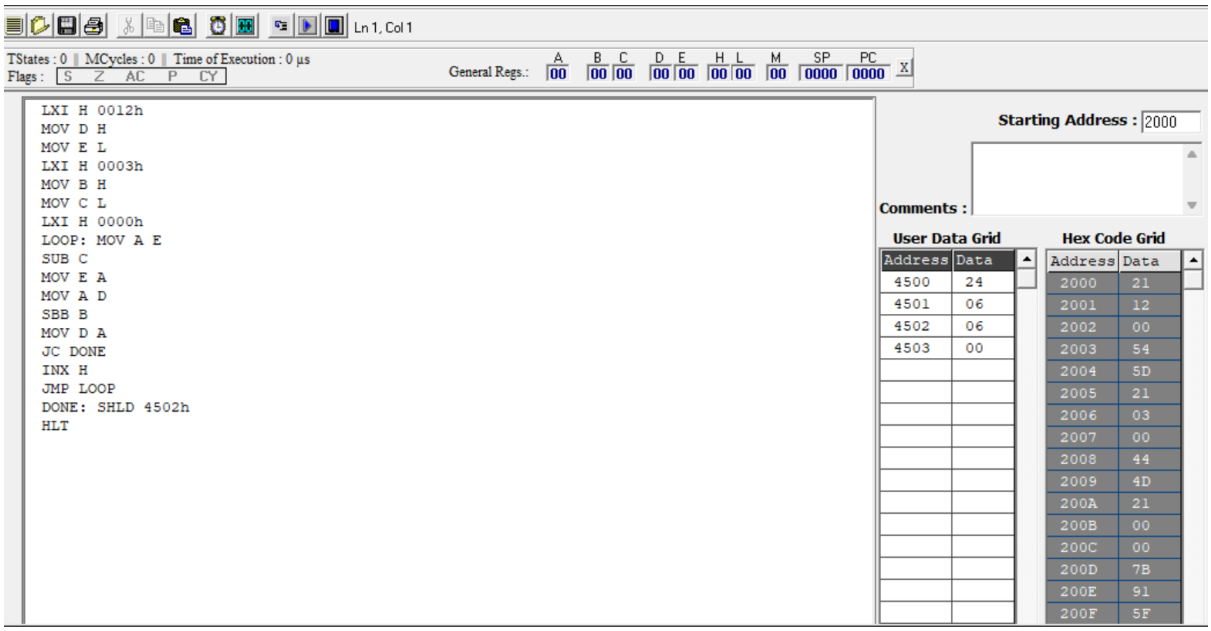
Input:

Input Address	Data
LXI H 0012h	LXI H 0012h
MOV D H	MOV D H

Output:

Output Address	Data
4502h	06h
4503h	00h

Output Screenshort:



Result:

Thus, the **8-bit addition and subtraction** and **16-bit multiplication and division** operations were successfully developed and verified using **8085 microprocessor assembly language programs**. The calculated results were obtained correctly and stored in the specified memory locations.

Student Reflection:

I learned how to perform **8-bit and 16-bit arithmetic operations using 8085 assembly language**. I understood the use of registers, register pairs, arithmetic instructions, loops, and memory locations. This experiment also helped me understand how multiplication can be performed using repeated addition and division using repeated subtraction. It improved my knowledge of **8085 programming and arithmetic operations**.